

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Ocean Gyres (Munk)

Lab 4

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Questions

Theory

1. State the necessary assumptions that are needed to study the depth-integrated gyre circulation.
2. Write down the definition of the barotropic stream function.
3. Write down the equation for the stream function when assuming boundary condition $\Psi_s(x_e) = 0$. State the name of the circulation that is described by the equation.
4. Describe the shape of the gyre.
5. State the additional assumptions for the Munk theory.
6. Describe the main changes in the shape of the stream function under Munk's theory.
7. Give the equation for the Munk solution.
8. Describe the (shape of the) wind stress (discuss the equation for the wind stress) and the geographical situation where this can commonly be applied.

Experiments**Sverdrup circulation**

1. Provide plots of the stream function for all your experiments at appropriate time steps.
2. What effect does $\beta = \partial f / \partial y$ have on the system?
3. What effect does the wind forcing have?
4. Provide cross sections (in x-direction) of the stream functions computed from your experiments.
5. From all your plots, discuss if the results resemble theory. Why / Why not?
6. Do the results improve if the size of the domain is increased?
7. What are the limitations of Sverdrup's model?

Munk's theory

1. Provide plots of the stream functions of your experiments.
2. Describe how the results have changed from Sverdrup circulation to Munk's theory.
3. Do theory and numerical solution correlate with each other?
4. Provide:
 - cross sections (in x-direction) of the stream functions computed from your experiments.
 - the difference of the cross sections of stream functions between theory and simulation results, and make sure that they also show the Munk thickness from theory.
5. From that, discuss the role of the friction coefficient for the shape.

1 Theory

i) In order to derive Sverdrup gyre circulation, one needs the following assumptions:

1. The Coriolis parameter is approximated using the β -plane:

$$f \approx f_0 + \beta y.$$

2. The flow needs to be large-scale, steady, and geostrophic. This is the same as stating a small Rossby number and negligible friction.
3. One needs some type of forcing, such as a latitudinally changing ten meter wind. The wind forcing is represented at the ocean surface as a wind stress, which drives Ekman pumping in the surface boundary layer. This Ekman pumping is then together with mass continuity responsible for the Ekman transport in the ocean.
4. The ocean basin is assumed to be laterally bounded, for example by having two continents on either side of the ocean.
5. We also assume a barotropic ocean, where there is no change with depth. This is however not completely true, since we also assume an Ekman layer, but since the size of this layer is much smaller than the domain, it is sufficient to say that the ocean is barotropic since $\delta_{\text{Ek}} \ll H$.

ii) The barotropic streamfunction is a useful quantity since it combines the zonal, u , and meridional, v , velocity into one parameter. The barotropic streamfunction is defined as

$$\frac{\partial \Psi}{\partial x} = vH, \quad \frac{\partial \Psi}{\partial y} = -uH, \quad (1)$$

where H is the mean depth of the domain.

iii, iv) The solution to this problem is the Sverdrup balance. Solving for an arbitrary ten meter wind stress, $\vec{\tau}$, one obtains the Sverdrup balance as

$$\beta vH = \frac{1}{\rho} \nabla \times \vec{\tau}. \quad (2)$$

Using the definition of the streamfunction above, one can derive the solution for the streamfunction itself. Integrating from an arbitrary point to the eastern boundary gives

$$\int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x} dx = \int_x^{x_{\text{East}}} vH dx = \Psi(x_{\text{East}}, y) - \Psi(x, y) \quad (3)$$

since we assume $\Psi(x_{\text{East}}, y) = 0$. The reason for assuming the eastern boundary to be zero and not the western, is because we also assume that $(\nabla \times \vec{\tau}) < 0$, i.e. that the wind travels clockwise around the basin. Since the water stress is in the same direction as the wind stress, we see that the water can only travel northward on the western boundary. Since the Sverdrup balance gives a net southward ocean flux in the basin center, $(\nabla \times \vec{\tau}) < 0$, we must have non-zero flux on the western boundary. Using the Sverdrup balance for the meridional velocity, one can obtain a complete solution as

$$\Psi(x, y) = -\frac{1}{\rho\beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau}) dx, \quad (4)$$

This gives an interesting shape of the gyre, where we see a net southward velocity in the latitudinal center of the domain, and less flux away from the center. Observing Equation 4, one can see that this yields positive streamfunction values. We will thus see a "parabolic-like" structure along the longitudinal direction, with the peak of the parabola in the east. We thus see zero flux close to the eastern boundary.

v) In traditional Munk theory we also add a Laplace friction term to the shallow water equations. We thus assume that we are given the set of shallow water equations as

$$\begin{cases} -fv = -g\frac{\partial\eta}{\partial x} + A\nabla_H^2 u + \frac{\tau^x}{\rho H} \end{cases} \quad (5a)$$

$$\begin{cases} fu = -g\frac{\partial\eta}{\partial y} + A\nabla_H^2 v + \frac{\tau^y}{\rho H} \end{cases} \quad (5b)$$

$$\begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \end{cases} \quad (5c)$$

where u and v are the horizontal velocity components in the x and y directions, η is the free surface height, H is the mean fluid depth, f is the Coriolis parameter, g is the acceleration due to gravity, A is the Munk viscosity parameter, τ is the stress, ρ is the density, and H is the mean depth.

However, in the Munk theory we have some additional assumptions:

1. We assume that the horizontal lateral friction is the only relevant viscosity, and can thus ignore the zonal viscosity $A\nabla_H^2 u$. We do this since when solving Equation 5 for the streamfunction we obtain

$$A\nabla^4\Psi - \beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla \times \vec{\tau}, \quad (6)$$

where $\frac{\partial\Psi}{\partial x} = vH$. Thus only the viscosity in the meridional velocity will affect the solution.

2. We assume that the solution in the interior remains like in the Sverdrup equation. We thus assume $\Psi \rightarrow \Psi_{\text{Sverdrup}}$ far away from the boundary.
3. Finally we assume that the streamfunction is bounded by the domain and $\Psi(x_{\text{East}}) = \Psi(x_{\text{West}}) = 0$.

vi) Thus the new Munk theory will also affect the shape of the theoretical solution. Previously we had a "parabolic-like" solution, with no closed form solution. However, with the Munk theory, we will theoretically see a western boundary current that returns the flow from the southward Sverdrup flux.

vii) Deriving the Munk solution with Equation 6 and the assumptions above one obtains

$$\Psi(x, y) = \Psi_{\text{Sverdrup}}(x, y) \left(1 - e^{-\frac{x-x_{\text{West}}}{2\delta_M}} \left[\cos\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_M}\right) + \frac{1}{\sqrt{3}}\sin\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_M}\right) \right] \right), \quad (7)$$

where Ψ_{Sverdrup} is the Sverdrup solution and δ_M is the Munk layer defined as

$$\delta_M \equiv \left(\frac{A}{\beta}\right)^{\frac{1}{3}}. \quad (8)$$

The Munk layer is the distance from the boundary to the latitude of largest northward meridional velocity.

viii) One can approximate the wind stress in the North Atlantic with

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin\left(\frac{\pi y}{L_y}\right), \quad (9)$$

where ρ_{air} is the air density, C_D is the drag coefficient, U_{10} is the wind speed measured at 10 meters above the ocean surface, and L_y is the meridional extent of the basin. The sinusoidal dependence represents easterly trade winds in the southern part of the North Atlantic and westerlies in the northern part. This τ is thus a pseudo accurate approximation of the wind stress in the North Atlantic.

2 Experiments

2.1 Sverdrup circulation

Neglecting the Munk theory and only focusing on the Sverdrup solution gives Figure 1–4. One can in Figure 1 see the result if only the f -plane is assumed. There one can see that the solution does not resemble the theory presented by Sverdrup. Here the solution is symmetric, which makes sense since there is no latitudinal asymmetry caused by the β -term. The observed solution in Figure 1 is thus simply a geostrophic solution to the wind stress, and thus symmetric in the domain.

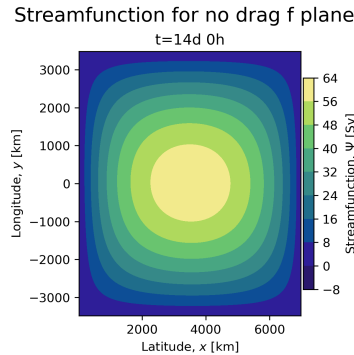


Figure 1: Streamfunction for the no-drag f -plane configuration. The circulation remains symmetric with no β -effect and without western intensification.

If one introduces the β -term one obtains results in Figure 2. Figure 2–5 uses a time averaged solution, since a steady state was never observed from the Sverdrup circulation. Thus without the time averaging (which can be seen in Figure 7) one can see a large discrepancy. However, when taking the time average, a more Sverdrup-like solution can be observed.

In Figure 2 one can observe more Sverdrup like solution compared to Figure 1, due to the asymmetric beta effect (stronger f northward and weaker southward) pushing of the solution towards the western boundary. The symmetric solution is thus broken and a better Sverdrup solution is obtained.

Varying the magnitude of the wind stress can be done by varying the ten meter wind as seen in Figure 2. There one can see the effect of a stronger wind stress on the Sverdrup circulation. It is clear that increasing the wind also increases the flux of the domain. This makes sense since the wind stress is the forcing that drives this circulation. Thus having more forcing should yield larger values of the streamfunction.

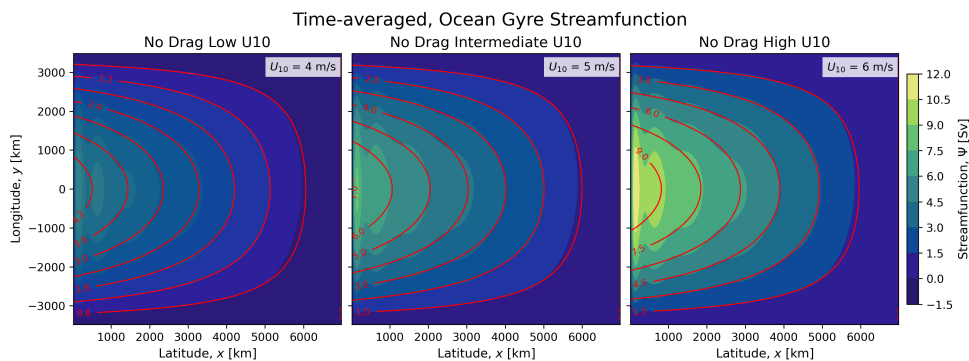


Figure 2: Comparison of streamfunction fields for different ten meter wind speeds. Increasing wind speeds shows that it increases the magnitude of the wind-driven ocean gyre. The contours show the model result, while the solid red lines showcase the Sverdrup theory. The time averaging was done between two days and two weeks for the model.

The cross sectional streamfunction for varying ten meter wind can be seen in Figure 3. There one can see that an increased wind shear (ten meter wind) yields a stronger meridional velocity. However, Figure 3 also shows the agreement to the theoretical result, where one can see good agreement for the time-averaging. Away from the western boundary the model and the theory align very good, but close to the western boundary they are very different. This discrepancy arises from the mass conservation of the numerical model. Since the theoretical Sverdrup is not in closed form, but goes from the top down to the bottom, there will be mass build up at this lower point. This was not a problem in the theoretical model since a steady state solution was assumed, but here we have time steps which highlight this difference. Thus due to mass conservation, the theoretical result will be impossible for the numerical model to mimic. However, far away from the boundary, the theoretical and numerical models still align very good.

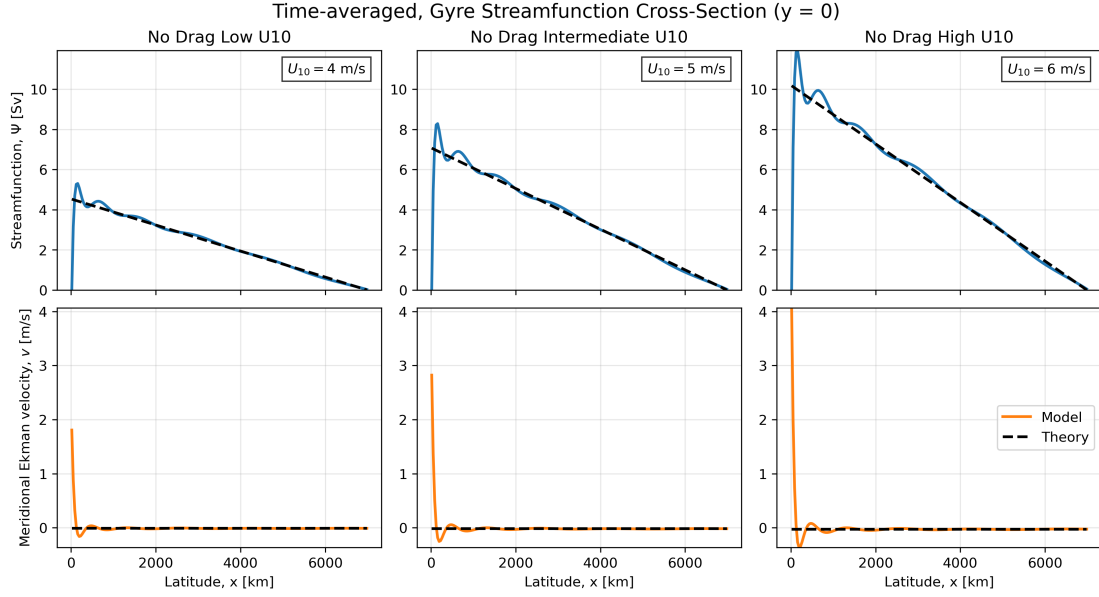


Figure 3: Zonal cross-sections of the streamfunction at the center of the domain for different ten meter winds. Solid lines show the numerical model results, while dashed lines show Sverdrup theory. A large difference between the model and the theory can be observed. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{EK} = 50$ m. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time averaging was done between two days and two weeks for the model.

The model ran on parameters resembling the North Atlantic, but changing this to a North Pacific scenario yielded Figure 4–5. There one can observe the effect if one increases the domain width, while keeping δx constant. Doing this shows the effect the domain width has on the solution. Observing Figure 4 one can see that we have a larger flux for the North Pacific compared to the North Atlantic. Observing Figure 5 one can see the effect of the domain increase on the accuracy. On the larger Pacific one can see that the solution is still bad in the western boundary, but away from the western boundary the solution is just as good as for the North Atlantic. One can also note that the increased domain pushed the peak and western boundary current closer to the western boundary.

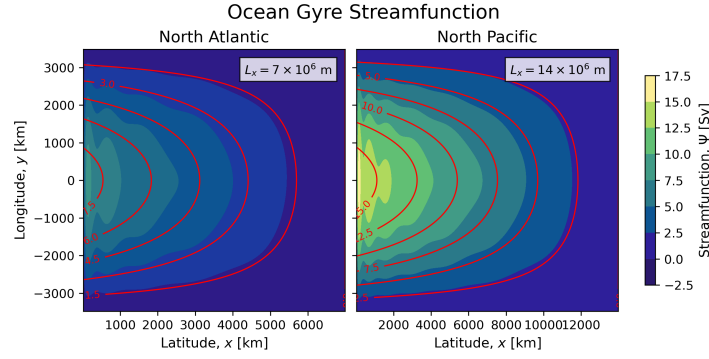


Figure 4: Comparison of streamfunction fields for a smaller domain (The North Atlantic) and a latitudinal larger domain (The Pacific). A larger circulation magnitude can be observed for the larger basin. The contours show the model result, while the solid red lines showcase the Sverdrup theory. The time averaging was done between two days and two weeks for the model.

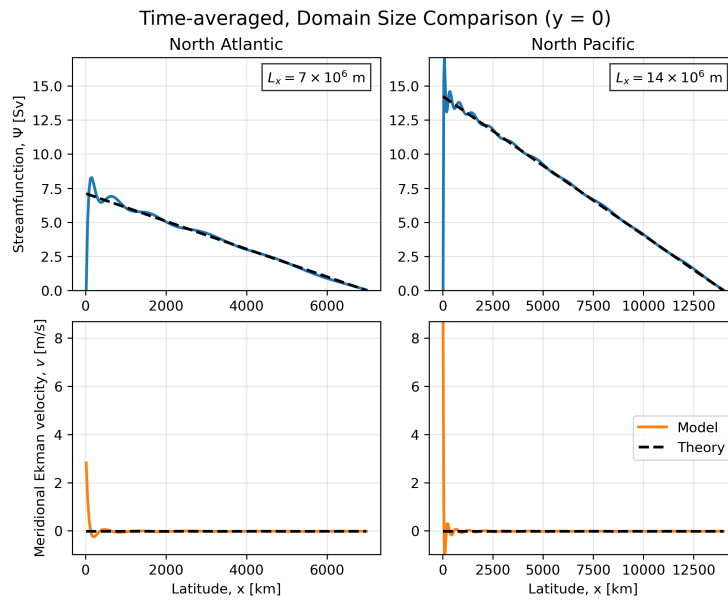


Figure 5: Comparison of streamfunction fields for a smaller domain (The North Atlantic) and a latitudinal larger domain (The Pacific). Solid lines show the numerical model results, while dashed lines show Sverdrup theory. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{EK} = 50$ m. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time averaging was done between two days and two weeks for the model.

Overall, clear limitations of the Sverdrup model can be observed due to the accumulation of mass along the western boundary. This leads to the formation of a strong western boundary current, which is not represented in the original Sverdrup theory. Therefore, some form of viscosity or friction must be included in the system to properly balance the boundary transport. Another clear limitation is the fact that the solution does not reach steady state (as seen in Figure 7), which forces one to do a time-averaging to get a realistic result.

2.2 Munk's theory

The difference by introducing Munk friction, compared to no friction, can be seen in Figure 6. There one can see multiple observations. Firstly one can see that the Munk theoretical results now include the western boundary current in both the numerical and theoretical solution. Therefore, the numerical solutions agrees much better with the theoretical one, compared to the Sverdrup solution. One can also see a small difference in a small tapering of the Munk numerical and theoretical solution, compared to the Sverdrup solution.

However, the largest distinction is the steady state convergence of the two different theories. Observing the Munk model, one can see that the numerical solution initially converges towards the theoretical one, and once the theoretical agreement is obtained the solution is steady. Almost the opposite is true for the Sverdrup solution. The agreement to the theoretical Sverdrup solution is okay after two days, but then becomes worse with time and does not reach a steady state. Furthermore, the mass build up creates a high that starts to propagate like a dispersive Rossby wave, which is very far from the theoretical solution. One can thus state that the steady state converging Munk solution is a big improvement from the Sverdrup solution. However, to make a fair comparison between the two models, the Munk solution in Figure 7–8 will also be time averaged, although this is not essential to yield realistic result.

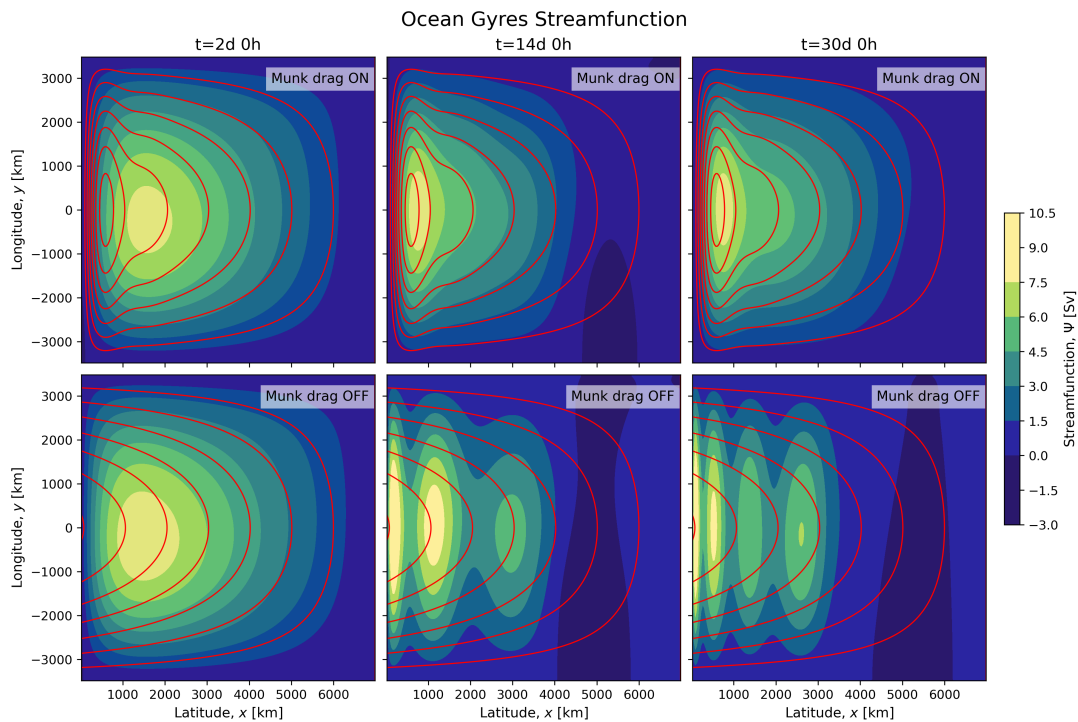


Figure 6: This figure shows the effect of the Munk friction, compared to the Sverdrup theory without friction. This is seen for different time-steps, to highlight the difference in steady state solutions. The Munk theory displays a proper convergence, whereas the Sverdrup does not converge to a steady state solution. The contours show the model result, while the solid red lines showcases the Munk and Sverdrup theory respectively.

Plots of the Munk theory can be seen in Figure 7–8. In Figure 7 one can see the effect of varying the drag coefficient A . Increasing the viscosity can be seen to do two major things: decreasing the speed of the western boundary current and moving the western boundary current further away from the western boundary. The decrease in speed for higher viscosity makes sense since there is more frictional "energy loss" in the system. The fact that the western boundary current is moved away from the boundary also makes sense since this increases the size of the Munk layer δ_M .

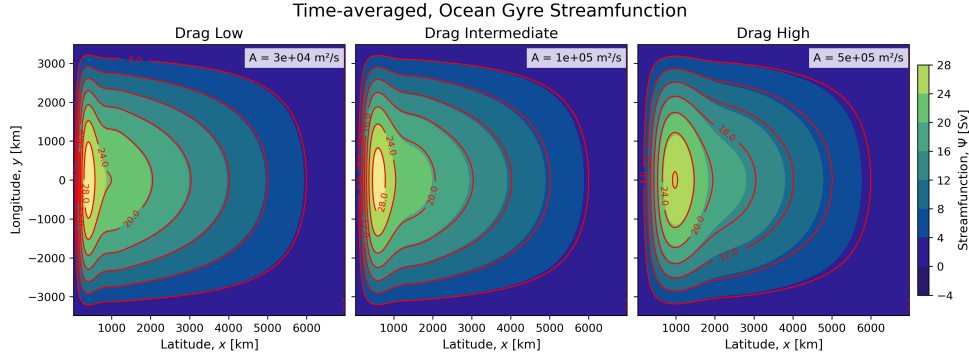


Figure 7: Streamfunction fields for different values of the meridional viscosity coefficient A , compared to Munk theory. The contours show the model result, while the solid red lines showcases the theory. The resolution of the Munk depth is $\frac{\delta M}{\Delta x} = [3.8, 4.9, 8.4]$, from left to right in the figure. One can observe a closer proximity of the theoretical result for the increasing A . The time-averaging was done between two days and two weeks.

The cross sectional streamfunction for varying viscosity wind can be seen in Figure 8. There one can see similar results to what was presented above: a stronger western boundary current for smaller viscosity together with a closer proximity to the western boundary. One can also observe that the smaller viscosity yields larger streamfunction fluxes. This makes sense since less viscosity means less energy loss to the system. Compared to Sverdrup solution one can see a much better result, with good agreement between the theory and the numerical model. Especially good result can be seen in the meridional velocity, which is almost spot on. Slightly worse results can be observed for the larger viscosity coefficient. There one can see a slight difference between the theoretical streamfunction and the numerical one.

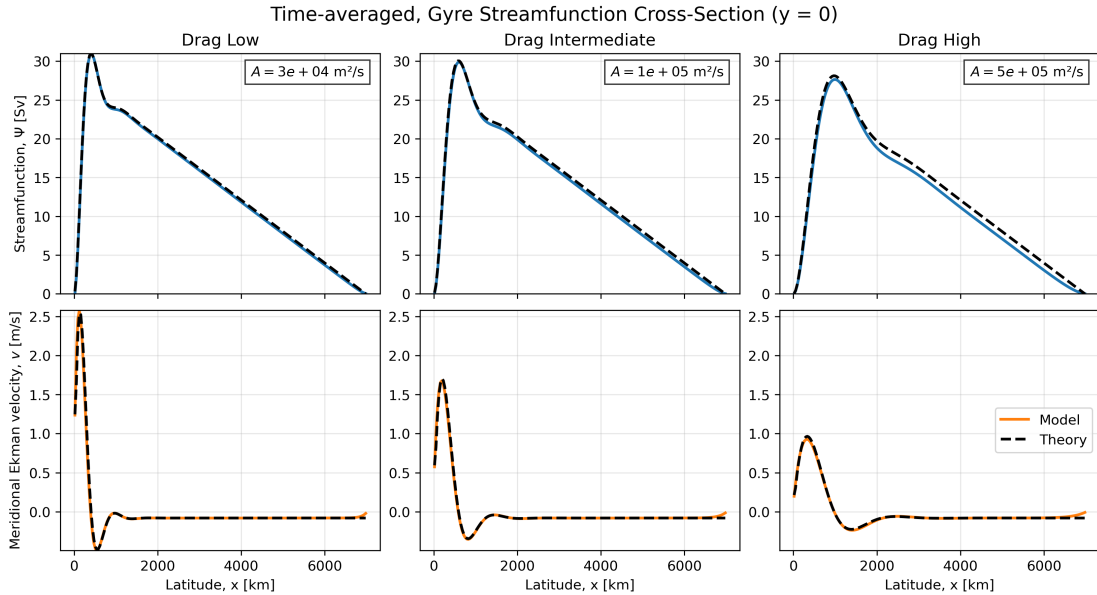


Figure 8: Zonal cross-sections of the streamfunction at the center of the domain for different values of the meridional viscosity coefficient A . Solid lines show the numerical model, while dashed lines represent Munk theory. Again the resolution of the Munk depth is $\frac{\delta M}{\Delta x} = [3.3, 4.9, 8.4]$, from left to right in the figure. Good agreement with theory can be seen. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{EK} = 50$ m. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time-averaging was done between two days and two weeks.

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`